

1(a)	actual width = image width ÷ magnification ; A $A = I \div M$ $M = I \div A$ $I = A \times M$ or magnification triangle working = width divided by 4275 ; e.g. 16 000 ÷ 4275 17 000 ÷ 4275 18 000 ÷ 4275 19 000 ÷ 4275 ; 3.7 (µm) 4.0 (µm) 4.2 (µm) 4.4 (µm) ; R answer if given to more than 1 dp or whole number	3
1(b)(i)	DNA – A / B / C ; cellulose – E ; phospholipid – A / C ; histone proteins – A / B ;	4
1(b)(ii)	chloroplast / mitochondrion ;	1
1(b)(iii)	nucleus ; A chloroplast / mitochondrion R nucleolus	1

Question	Answer	Marks
1(c)	<i>any two from</i> 1 (section at) high <u>resolution</u> ; A suggestion of a correct value of resolution for a TEM 2 any named structure visible in Fig. 1.1 that can, only be seen in a TEM / not be seen in a photomicrograph ; e.g. internal structure of chloroplasts / thylakoid(s) / grana e.g. internal structure of mitochondria / cristae 3 high magnification / higher magnification (magnification > 1000 / higher than with light microscope) ; <i>in context of higher than light microscope</i> 4 (very) thin ; 5 2D / no surface contours / no surface features / AW ; A not 3D	2

Question	Answer	Marks
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